

PROJECT 05: STUDENT PERFORMANCE & ACADEMIC ANALYTICS

Empirical Exploratory Analysis of Socio-Demographic & Behavioral Drivers

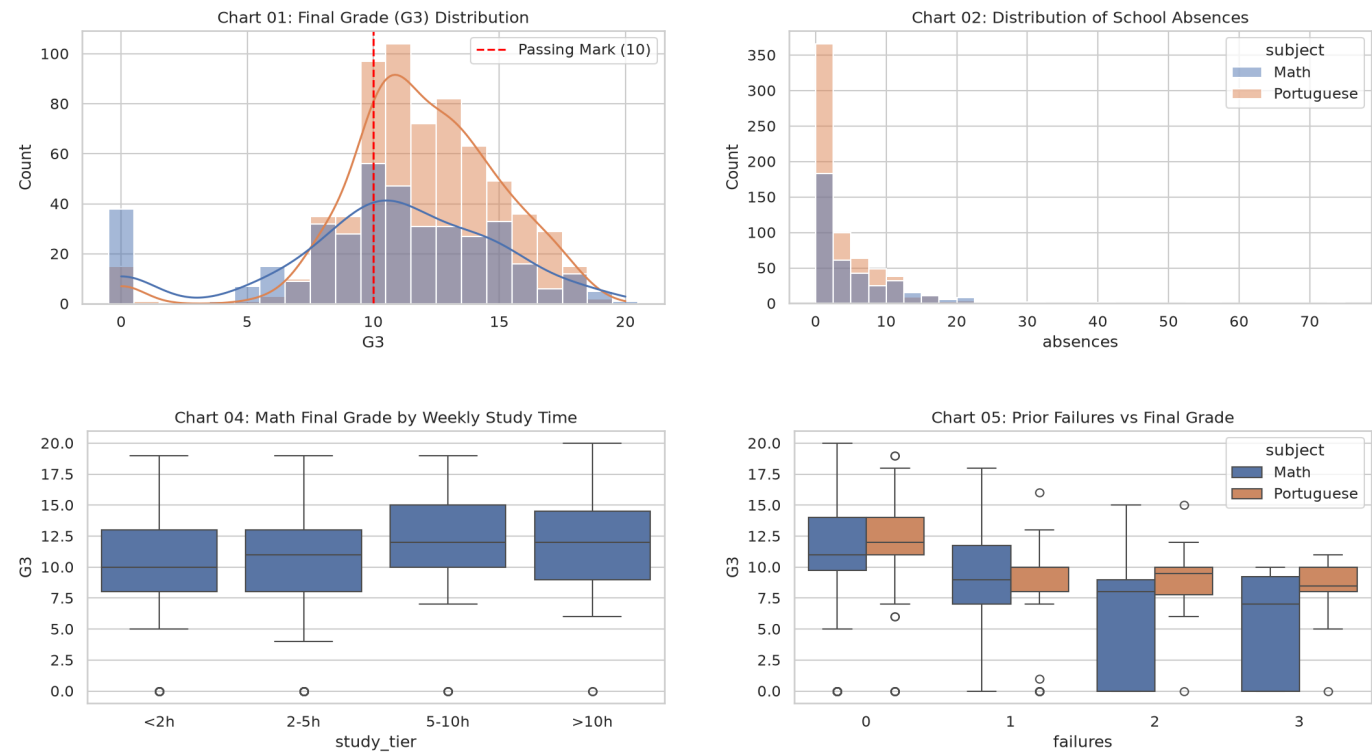
Author: Kushagra Pratap Singh | UCI Student Performance Dataset

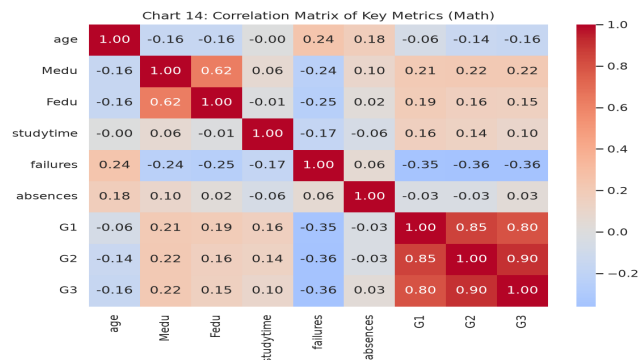
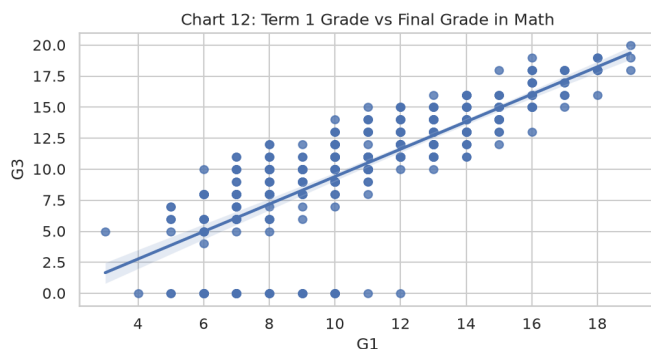
1. Executive Summary & Problem Formulation

This report presents an end-to-end exploratory analysis of secondary student performance in Mathematics (395 records) and Portuguese Language (649 records). By analyzing 33 demographic, lifestyle, behavioral, and academic variables, we evaluate predictors of student success, address the attendance paradox, and identify critical early intervention milestones.

Evaluation Metric	Math (N=395)	Portuguese (N=649)	Key Comparative Insight
Final Grade Mean (Std)	10.42 (4.58) / 20	11.91 (3.23) / 20	Math exhibits 41% higher variance and lower mastery
Passing Rate ($\geq 10/20$)	67.09% (265 Passed)	84.59% (549 Passed)	Language achievement exceeds Math by 17.5%
Zero-Grade Exam Dropouts	38 Students (9.62%)	15 Students (2.31%)	Significant exam abandonment cluster in Math
Term Correlation: G1 to G3	$r = +0.801$ (Strong)	$r = +0.826$ (Strong)	Term 1 performance strongly anchors final results
Term Correlation: G2 to G3	$r = +0.905$ (Very Strong)	$r = +0.919$ (Very Strong)	Academic trajectory is locked in by mid-year
Mother's Education (Medu)	$r = +0.217$ (Positive)	$r = +0.240$ (Positive)	Mother's education is primary socio-economic driver

2. Visual Analysis & Empirical Evidence





3. Answers to the 10 Core Research Questions

Q1: Distribution of Final Grades: Math displays a bimodal distribution (mean 10.42) with a 9.6% dropout spike at 0, while Portuguese shows higher average mastery (mean 11.91) and an 84.6% pass rate.

Q2: Study Time Elasticity: Grades rise from 10.05 for <2h/week to 11.40 for 5-10h/week. Above 10 hours, grades plateau (11.26), showing diminishing returns where study technique matters more than hours.

Q3: The Attendance Paradox: Raw correlation is flat (+0.034) because dropouts accumulated 0 absences after abandoning exams early. Among active test-takers, absences show a clear negative correlation (-0.213).

Q4: Parental Education Impact: Mother's education ($r = +0.217$) exerts a stronger influence than Father's ($r = +0.152$). Dual-college households average 12.50-12.96, compared to 9.95 for primary-educated households.

Q5: Family Background Drivers: Maternal employment in healthcare (12.15) or teaching (11.05) correlates with higher marks than at-home roles (9.15). Family relationship quality (famrel) has a positive association.

Q6: Historical Academic Performance: Prior grades are the single strongest predictor: G1 correlates at $r = 0.801$, and G2 at $r = 0.905$. Each prior subject failure reduces final achievement by an average of 2.19 grade points.

Q7: Extracurricular Activities: Extracurriculars have negligible direct linear grade impact (10.49 vs 10.34 in Math), but correlate strongly with higher education aspirations (>90%) and positive social integration.

Q8: Gender Differences Across Curricula: Male students score higher in Mathematics (10.91 vs 9.97), while female students outperform males in Portuguese Language (12.25 vs 11.41), reflecting subject-specific engagement.

Q9: Strongest Predictors of Final Grades: Ranked in order: (1) Second Period Grade G2 ($r = +0.905$), (2) First Period Grade G1 ($r = +0.801$), (3) Past Failures ($r = -0.360$), (4) Mother's Education ($r = +0.217$).

Q10: Outlier & Dropout Identification: 38 Math students scored zero due to late-term course abandonment despite passing in Term 1 (mean G1 = 9.0). On the high end, 18 students scored $\geq 18/20$ (0 failures, study time tier 3).

4. Actionable Institutional Recommendations

- **Term 1 Early Warning System:** Automatically flag students scoring G1 ≤ 9 or with prior failures for academic coaching before Term 2 performance locks in.
- **Attendance Disengagement Interventions:** Monitor sudden consecutive absences between Term 1 and Term 2 to intervene before students drop out or abandon exams.
- **Restructure Remedial Support:** Shift institutional support (schoolsup) from late-stage corrective measures into early small-group mastery workshops.
- **Study Skills & Active Recall Workshops:** Train students on active recall, spaced repetition, and problem-solving strategies rather than unguided study volume beyond 10 hours/week.
- **Targeted First-Generation Support:** Establish structured study halls and peer mentorship networks for students whose parents did not attend higher education.